DIFFERENTIAL EQUATIONS AND APPLLICATIONS

ALEKS ENGINEERING VIBES /TEAM GERAMA – 2025

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PART 1

1) **Prove whether the** $f(x,y) = x^4 - x^3y$ **function is homogeneous** **[2marks -2016]**

2) Find the value of λ, for which the differential equation

$$(xy^2 + \lambda x^2 y)dx + (x+y)x^2 dy = 0 \text{ is } \textbf{exact}$$

Solve the equation for this value of λ. **[6marks-2016]**

3) **Solve using separation of variables the differential equation** **[5marks- 2016]**

$$\frac{dy}{dx} - 2y\tan x = y^2 \tan x$$

4) The integrating factor of the differential equation $(x-2)\frac{dy}{dx} - y = (x-2)^3$ I s

a) $\frac{1}{x-3}$ b) $x-2$ c) $x-3$ d) $\frac{1}{x-2}$

5) The **integrating factor** to make the differential equation $x^2y\,dx - (x^3+y^3)dy = 0$ exact is:

(a) y^4 (b) $-y^{-4}$ (c) x^2y^2 (d) x^4

6) The differential of the form $\frac{dy}{dx} + P(x)y = Q(x)y^n$ is called:

a) First Order Linear equation

b) Exact differential equation

c) Separation of variables

d) Bernoulli's equation

7) Consider the differential equation $L\frac{d^2q}{dt^2} + R\frac{dq}{dt} + \frac{q}{c} = Esin(\omega t)$. Identify all the llowing terms

a. Dependent variable = ...

c. Independent variable = ...

b. Degree = ...

d. Constants = ... [0.5 each]

8) Consider the differential equation $\frac{d^2y}{dt^2} + \omega_0^2 y = F_0 \sin \omega t + A$. Identify the Dependent and Independent variable and the Constants.

i. Dependent variable .....[0.5 Ma

ii. Independent variables .....[0.5 Ma

iii. Constants ..... [2 Ma

[3Marks – 2021]

9) Consider the differential equation $\frac{d^2y}{dt^2} + \omega_0^2 y = F_0 sin\omega t + A$. Identify all the following terms.

i. Dependent variable ..... $[0.5mark]$

ii. Degree ..... $[0.5mark]$

iii. Independent variables ..... $[2marks]$

[2019, 2018]

10) Write down the **general form** of the following differential equations

a. Linear first order : ..... [0.5 mark]

b. Exact differential equation : ..... [0.5 mark]

c. Bernoulli's differential equation : ..... [0.5 mark]

d. Cauchy- Euler's differential equation : .....

e. Linear nth order differential equation: .....

..... [1 mark]

f. Lagrange's Equation : ..... [0.5mark]

[2017]

11) Investigate whether the function $h(x,y) = \frac{y^2+xy}{\sqrt{y^4-x^4}}$ is homogeneous. [2 marks -2017]

12) Solve the exact differential equation

$$(4e^{2x} + 2xy - y^2)dx + (x-y)^2 dy = 0$$

[5marks -2017]

13) Using the substitution $y = vx$ solve the differential equation

$$(2xy + x^2)\frac{dy}{dx} = 3y^2 + 2xy$$

[5 marks]

PART 2

(Midsem PQs)

14) Find the differential equation corresponding to the family of curves $y = c(x-c)^2$, where c is an arbitrary constant. [4 marks]

15) Write the **general form** for the following differential equations

a) Exact differential equation [1 mark]

b) Linear first order differential equation [1 mark]

c) Bernoulli's equation. [1 mark]

16) Use the appropriate method to solve the following differential equations

a) $x^2y - x^3\frac{dy}{dx} = y^4 cosx$ [5 marks]

b) $\frac{dy}{dx} = e^{x-y} + x^2e^{-y}$ [5 marks]

PART 3

17) Form differential equation of the family of curves $y^2 = 4a(x+a)$ where a is an arbitrary constant.

[2021]

18) Solve the differential equation $\frac{dy}{dx} = \frac{x+2y-1}{2x-y+3}$ (8 Marks)

[2021]

19) The variables x and y are related by the differential equation $\frac{dy}{dx} = \frac{1}{5}x\sqrt{y}\sin(\frac{x}{3})$. Given that $y = 100$ when $x = 0$, find the value of y when $x = 25$.

20) Using the transformation $y = \frac{v}{x}$ solve the differential equation

$$(y - xy^2) = (x - x^2y)\frac{dy}{dx}.$$ [7marks]

21) **Form a differential equation from**

$$y = ae^x + be^{2x}$$ [4 marks]

[2019, 2017,2018]

PART 4

22) **Find the order and degree of** $\frac{d^2y}{dx^2} + \sqrt{x + \left(\frac{dy}{dx}\right)^3} = 0$ [2 marks]

[2017, 2018

23) **Solve the exact differential equation**

a.

$$2xe^y dx + (x^2e^y + cosy)dy = 0$$ [5marks]

[2017]

b. $$(4e^{2x} + 2xy - y^2)dx + (x^2 - 2xy + y^2)dy = 0$$

[2018]

c. $$2x^2\frac{dy}{dx} + 4xy = 3\sin x, \quad y(2\pi) = 0 \quad (5 \text{ Marks})$$

[2020]

d. $$2x^2\frac{dy}{dx} + 4xy = 3sinx, \quad y(2\pi) = 0. \quad [6\text{marks}]$$

[2018]

e. Find the value of λ, for which the differential equation

$$(xy^2 + \lambda x^2 y)dx + (x + y)x^2 dy = 0 \text{ is exact}$$

Solve the equation for this value of λ. [5marks]

24) **Solve using separation of variables**

[2017]

$$x(y - 3)\frac{dy}{dx} = 4y$$

PART 5 -Assignment Questions

25) Solve the following differential equations:

a) $(x+4)y\frac{dy}{dx} = x(y-3)$

b) $y\frac{dy}{dx} = a\left(y^2 + \frac{dy}{dx}\right)$

c) $\frac{dy}{dx} + 3x^2y = x^5e^{x^3}$

d) $(x+a)^3\frac{dy}{dx} - 3y = (x+a)^5$

e) $(x^3y^2 + xy)dx = dy$

f) $\frac{1}{y^2}\frac{dy}{dx} - \frac{1}{y} = 2xe^{-x}$

g) $\frac{dy}{dx} = \frac{y^2+y}{x^2-x}$; $x = 5, y = \frac{3}{2}$

26) Investigate whether the following functions are homogeneous:

a) $f(x,y) = \frac{x\sqrt{y^2+x^2}+y^2}{xy}$

b) $f(x,y) = \frac{x^2+y}{xy}$

c) $f(x.y) = \frac{x\sin\left(\frac{x}{y}\right) - y\cos\left(\frac{x}{y}\right)}{y}$

27) Solve the following differential equations:

a) $x\frac{d^2y}{dx^2} + 2\frac{dy}{dx} + 4xy = 8x^2 + 16$

b) $y^{ll} + 5y^{l} + 4y = 3 - 2x$

c) $y^{ll} - 3y^{l} + y = xe^{2x} + 1.$

d) $y^{ll} + 2y^{l} + y = e^{-x}\ln x$

e) $x\frac{dy}{dx} + 2\frac{dy}{dx} + 4xy = 8x^2 + 16$

PART 6

28) ; Classify the following as either **linear** or **non-linear** differential equation

a. $\frac{d^2y}{dx^2} + \frac{dy}{dx} + y^2 = sinx$ ..... [0.5 mark]

b. $2\frac{d^2x}{dt^2} - \frac{dx}{dt} - 3x = f(t)$ ..... [0.5 mark]

(Mid 2019)

29) Write the *complementary solution* using D-operator method.

$$\frac{d^4y}{dx^4} - m^4y = 0$$

(Mid 2019)

30) The general solution of $(D^2 - m^2)y = 0$ is:

(a) $y = (c_1 + c_2x)e^{mx}$ (b) $c_1e^{mx} + c_2e^{-mx}$ (c) $y = c_1 \sin mx + c_2 \cos mx$
(d) None of these

(Mid 2019)

32) (a) Find a particular solution of the equation $y'' + y = \tan x$ (5 Marks)

33) Use the D-operator method to solve the differential equation $\frac{d^2y}{dx^2} - 2k\frac{dy}{dx} + k^2y = e^x$. [4 marks]

34) Using the Wronskian method of variation of parameters, solve

$$y'' + y = tanx$$ [5 marks]

35)

. Find the Wronskian of $f_1 = x, \quad f_2 = x + x^2, \quad f_3 = 2x - x^2$

36)

(a) Given that $x \neq 0$ and $x\frac{d^2y}{dx^2} + 2\frac{dy}{dx} + 4xy = 8x^2 + 16$

Show that if $z = xy$ then the above equation becomes

$$\frac{d^2z}{dx^2} + 4z = 8x^2 + 16 \qquad \star$$

Find y (by first solving $\star$ and then replace z by xy) in terms of x, given that $y = 0$ and $\frac{dy}{dx} = -2$ when $x = \frac{\pi}{2}$. (8 Marks)

37) . The general solution of $(D^2 - m^2)y = 0$ is:

(a) $y = (c_1 + c_2x)e^{mx}$

(b) $y = c_1e^{mx} + c_2e^{-mx}$

c) $y = c_1 \sin mx + c_2 \cos mx$

(d) None of these

38) The solution of the differential equation $\frac{d^2y}{dx^2} - 4\frac{dy}{dx} + 4y = 0$ is :

(a) $y = (A_1 + A_2x)e^{2x}$

(b) $y = A_1e^{2x} + A_2e^{-2x}$

(c) $y = (A_1 + A_2x)e^{-2x}$

(d) $y = (A_1 - A_2x)e^{-2x}$

39)

Solve the following differential equations:

a) $x\frac{d^2y}{dx^2} + 2\frac{dy}{dx} + 4xy = 8x^2 + 16$

b) $y^{II} + 5y^{I} + 4y = 3 - 2x$

c) $y^{II} - 3y^{I} + y = xe^{2x} + 1.$

40)

. *Reduce* the Cauchy-Euler differential equation to D-operator form (**without finding the solution**)

a) $x^2y'' - 3xy' + 5y = x^2 sin(Inx)$, b) $x^4y''' - 2x^3y'' - x^2y' + xy = 1$

41) Using the Wronskian method of variation of parameters, solve

$$y'' + y = tanx$$ [5 marks]

44) (b) Find the Wronskian of $y_1 = x$, $y_2 = x + x^2$, $y_3 = 2x - x^2$.

QUIZ 1

1)

Using the substitution $y = vx$ solve the differential equation

$$(2xy + x^2)\frac{dy}{dx} = 3y^2 + 2xy$$ [5 marks]

QUIZ 2

1)

Solve the exact differential equation $(4e^{2x} + 2xy - y^2)dx + (x^2 - 2xy + y^2)dy = 0$

QUIZ 3

1)

Determine the degree of the following differential equations;

$$\left[\frac{d^2y}{dx^2} + \frac{dy}{dx}\right]^{\frac{1}{5}} = \left[\frac{dy}{dx} + \sin^2 x\right]$$

2)

Which of the following differential equations is linear:

(a) $\frac{dy}{dx} + x^2y = \sin y$

(b) $(1+y)\frac{dy}{dx} + \sin x = 0$

(c) $\frac{dy}{dx} - x^2y = \sin x$

(d) $\frac{dy}{dx} + y(y+x) = x^2$

3)

$$x\frac{dy}{dx} + 2\frac{dy}{dx} + 4xy = 8x^2 + 16$$

4)

$$\dot{y}^{ll} + 2\dot{y}^{l} + \dot{y} = e^{-x}\ln x$$

Exercises

Solve each of the differential equations in Exercises 1–14.

1. $4xy\,dx + (x^2 + 1)\,dy = 0.$
2. $(xy + 2x + y + 2)\,dx + (x^2 + 2x)\,dy = 0.$
3. $2r(s^2 + 1)\,dr + (r^4 + 1)\,ds = 0.$
4. $\csc y\,dx + \sec x\,dy = 0.$
5. $\tan\theta\,dr + 2r\,d\theta = 0.$
6. $(e^v + 1)\cos u\,du + e^v(\sin u + 1)\,dv = 0.$
7. $(x + 4)(y^2 + 1)\,dx + y(x^2 + 3x + 2)\,dy = 0.$
8. $(x + y)\,dx - x\,dy = 0.$
9. $(2xy + 3y^2)\,dx - (2xy + x^2)\,dy = 0.$
10. $v^3\,du + (u^3 - uv^2)\,dv = 0.$
11. $\left(x\tan\frac{y}{x} + y\right)dx - x\,dy = 0.$
12. $(2s^2 + 2st + t^2)\,ds + (s^2 + 2st - t^2)\,dt = 0.$
13. $(x^3 + y^2\sqrt{x^2 + y^2})\,dx - xy\sqrt{x^2 + y^2}\,dy = 0.$
14. $(\sqrt{x + y} + \sqrt{x - y})\,dx + (\sqrt{x - y} - \sqrt{x + y})\,dy = 0.$

Solve the initial-value problems in Exercises 15–20.

15. $(y + 2)\,dx + y(x + 4)\,dy = 0, \qquad y(-3) = -1.$
16. $8\cos^2 y\,dx + \csc^2 x\,dy = 0, \qquad y\left(\frac{\pi}{12}\right) = \frac{\pi}{4}.$
17. $(3x + 8)(y^2 + 4)\,dx - 4y(x^2 + 5x + 6)\,dy = 0, \qquad y(1) = 2.$

Exercises

Find the general solution of each of the differential equations in Exercises 1–24.

1. $\frac{d^2y}{dx^2} - 5\frac{dy}{dx} + 6y = 0.$
2. $\frac{d^2y}{dx^2} - 2\frac{dy}{dx} - 3y = 0.$
3. $4\frac{d^2y}{dx^2} - 12\frac{dy}{dx} + 5y = 0.$
4. $3\frac{d^2y}{dx^2} - 14\frac{dy}{dx} - 5y = 0.$
5. $\frac{d^3y}{dx^3} - 3\frac{d^2y}{dx^2} - \frac{dy}{dx} + 3y = 0.$
6. $\frac{d^3y}{dx^3} - 6\frac{d^2y}{dx^2} + 5\frac{dy}{dx} + 12y = 0.$
7. $\frac{d^2y}{dx^2} - 8\frac{dy}{dx} + 16y = 0.$
8. $4\frac{d^2y}{dx^2} + 4\frac{dy}{dx} + y = 0.$
9. $\frac{d^2y}{dx^2} - 4\frac{dy}{dx} + 13y = 0.$
10. $\frac{d^2y}{dx^2} + 6\frac{dy}{dx} + 25y = 0.$
11. $\frac{d^2y}{dx^2} + 9y = 0.$
12. $4\frac{d^2y}{dx^2} + y = 0.$
13. $\frac{d^3y}{dx^3} - 5\frac{d^2y}{dx^2} + 7\frac{dy}{dx} - 3y = 0.$
14. $4\frac{d^3y}{dx^3} + 4\frac{d^2y}{dx^2} - 7\frac{dy}{dx} + 2y = 0.$
15. $\frac{d^3y}{dx^3} - 6\frac{d^2y}{dx^2} + 12\frac{dy}{dx} - 8y = 0.$
16. $\frac{d^3y}{dx^3} + 4\frac{d^2y}{dx^2} + 5\frac{dy}{dx} + 6y = 0.$
17. $\frac{d^3y}{dx^3} - \frac{d^2y}{dx^2} + \frac{dy}{dx} - y = 0.$
18. $\frac{d^4y}{dx^4} + 8\frac{d^2y}{dx^2} + 16y = 0.$
19. $\frac{d^5y}{dx^5} - 2\frac{d^4y}{dx^4} + \frac{d^3y}{dx^3} = 0.$
20. $\frac{d^4y}{dx^4} - \frac{d^3y}{dx^3} - 3\frac{d^2y}{dx^2} + \frac{dy}{dx} + 2y = 0.$
21. $\frac{d^4y}{dx^4} - 3\frac{d^3y}{dx^3} - 2\frac{d^2y}{dx^2} + 2\frac{dy}{dx} + 12y = 0.$
22. $\frac{d^4y}{dx^4} + 6\frac{d^3y}{dx^3} + 15\frac{d^2y}{dx^2} + 20\frac{dy}{dx} + 12y = 0.$
23. $\frac{d^4y}{dx^4} + y = 0.$
24. $\frac{d^5y}{dx^5} = 0.$

PART 7

45) Find the **inverse Laplace** transform of the following

$\frac{3s-8}{4s^2-25}$ b) $\frac{s^2}{s^2+5}$ c) d) $\frac{4}{s^3}$ [8 marks]

46) Evaluate the following

$\frac{1}{D^2}e^{-x}$ b) $\frac{1}{(D+1)(D-1)}cosx$ c) $\frac{1}{(D-2)(D-3)}e^{2x}$ [6 marks]

2018, Midsem

47) Find the Laplace transform of the following

ı) $F(t) = e^{-t}cos^2t$ b) $\frac{1-cos4t}{3}$ c) $y'''(t)$ d) $e^{-2t}(3cos6t - 5sin6t + t^3)$

2018, Midsem

48) Use Laplace transform to solve the initial value problem

$y'' - y = 6\cos t,\ \ y(0) = 0,\ y'(0) = 4$ (6 Marks)

2021,Sec C

49) (a) Evaluate $\mathcal{L}^{-1}[\frac{2s+3}{(s-5)(s+4)}]$ (5 Marks)

2021, Sec C

50) Using laplace transform, solve the differential equation,

$y'' - 3y' + 2y = e^{-t} \quad y(0) = 3\ \ y'(0) = 4$ 10 Marks

2021, Sec C

51) Use the D-operator method to solve the differential equation $\frac{d^2y}{dx^2} - 2k\frac{dy}{dx} + k^2y = e^x$. [4 marks]

2016, Midsem

52) Find the Laplace transform of the following

a) $e^{-t}cos^2t$ [2 marks]

b) $\frac{1-sin4t}{3}$ [2marks]

53) Find the **inverse** L.T of the following

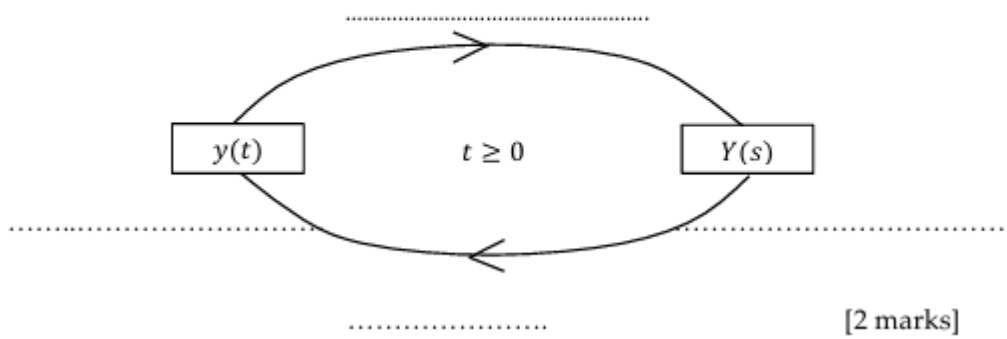
a) $\frac{s^2+2s+6}{s^3}$ [3 marks]

b) $\frac{1}{2s(s-3)}$ [3 marks]

52,532016, Midsem

54)

(a) Complete the diagram below .



[2 marks]

2016, 2017

55) The Laplace transform of $L\{t^3e^{-4t}\}$ is

a) $\frac{6}{(s-4)^4}$ b) $\frac{3!}{(s-4)^3}$ c) $\frac{3!}{(s+4)^3}$ d) $\frac{3!}{(s+4)^4}$

2017,

56) The partial fraction decomposition of $\frac{s+4}{(s-1)^2(s^2+4)}$ is

a) $\frac{A}{(s-1)^2}+\frac{Bs+C}{s^2+4}$

b) $\frac{A}{(s-1)}+\frac{B}{(s-1)^2}+\frac{Cs}{s^2+4}$

c) $\frac{A}{(s-1)}+\frac{B}{(s-1)^2}+\frac{Cs+D}{s^2+4}$

d) $\frac{A}{(s-1)}+\frac{B}{(s-1)^2}+\frac{C}{s^2+4}$

2017

57) Write the **Laplace transform** of the following functions:

a) $tcosh4t =$

.....

.....

.....

.....[1 mark]

b) $(t+1)U(t-1) =$

.....

.....

.....

..... [1 mark]

c) $e^{2t}cost =$

.....

.....

.....

..... [1 mark]

d) $[r(t)] =$..... .....

.....[0.5mark]

e) $\frac{d^6y}{dt^6} =$.....

.....

.....[1marks]

2017

58) Write the **Inverse Laplace Transform** of the following:

a) $\frac{e^{-\pi s}}{(s+3)}$ = ...

.....

.....

.....

.....

..... [2 mark]

b) $\frac{3s-8}{4s^2+25}$ = ...

.....

.....

.....

.....

....[2 mark]

d) $\frac{s-3}{s^2+5}$ = ...

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.....

.....

..... [2 mark]

c) $\frac{\sqrt{2}}{(s+3)^2-2}$ =.....

.....

.....

.....

....

..... [2 mark]

2018.

59) Evaluate the following

a. $\frac{1}{(D-2)(D-3)}e^{2x}$

.....

b. $\frac{1}{(D+1)(D-1)}cos(x)$

.....

60) c. Find y_p in

$$(D^3 - D^2 - 6D)y = x^2 + 1$$

.....

d. Find y_p in $D^2y - 2Dy + y = cos3x$

.....

.....

59, 58, 2018, 2Marks each

PART 8

61) Consider the $L-R$ series circuit given by $Ri + L\frac{di}{dt} = E$. Find the current at the **start** of flow. Support it with a diagram.

....

....

2018, 4marks

62) Solve the initial value problem

$y'' + 4y' + 13y = \delta(t-\pi)$, where $y(0) = 2$ and $y'(0) = 1$. [5 marks]

2018

63) . a) Solve the initial value problem $y'' + 4y' + 13y = \delta(t-\pi)$, where $y(0) = 2$ and $y'(0) = 1$. [5 marks]

2016

64) a) Solve the initial value problem $y'' - y = F(t)$ if $y(0) = 0$ and $y'(0) = 0$

where $F(t) = \begin{cases} 1, & 0 \le t < 1 \\ -1, & t \ge 1 \end{cases}$ [7 marks]

2016

65) Find the solution of $\frac{di^2}{dt^2} + \frac{R}{L}\frac{di}{dt} + \frac{i}{LC} = 0$ where $R^2C = 4L$ with R, L, C being constants. [3marks]

2017, Midsem

PART 9

66)

. a) Define *Dirac delta function* mathematically.

.....

.....

..... [1 mark]

67) Write the **Laplace transform** of the following

$(t+3)U(t-2)$ b) $2t^2e^{-3t}cosht$ c) $\frac{e^{5t-3}}{e^{\pi-4t}}$

68)

Express in terms of unit step

$$F(t) = \begin{cases} 2, & 0 \le t < 1 \\ 2e^{t-1}, & t \ge 1 \end{cases}$$ [3 marks]

69)) Solve the **Lagrange equation** $y^2p - xyq = x(z-2y)$ where $p = \frac{\partial z}{\partial x}$ and $q = \frac{\partial z}{\partial y}$

[4marks]

70) Solve the PDE by direct integration

$$\frac{\partial^3 z}{\partial x^2 \partial y} = \cos(2x+3y)$$ [3 marks]

71) Solve the initial value problem

$y'' + 4y' + 13y = \delta(t-\pi)$, where $y(0) = 2$ and $y'(0) = 1$. [5 marks]

72)

Form PDE from the equation $2z = \frac{x^2}{a^2} + \frac{y^2}{b^2}$ where a and b are arbitrary constants

73) . Evaluate $\int_{1}^{3}\left(t^2+4\right)\delta\left(t-2\right)dt$

(a) 2 (b) 4 (c) 6 (d) 8

2018,

74)

12. Fill in with the required second order partial differential equation (PDE)

Name of PDE	Equation of PDE
Laplace Equation in 3-D	
	$V_{xx} = RCV_t$ and $i_{xx} = RCi_t$

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Diffusion Equation in 1-D	
Poisson Equation in 3-D	
	$V_{xx} = LCV_{tt}$ and $i_{xx} = LCi_{tt}$
Wave Equation in 2-D	
	$p_x = \frac{1}{\mathcal{R}}\nabla^2 u, p_y = \frac{1}{\mathcal{R}}\nabla^2 v, u_x + v_y = 0$
Helmohltz Equation	

[0.5 marks each]

2018,

75) [2marks]

Express as unit step function

$$F(t) = \begin{cases} 0 & ,0 < t < 1 \\ t-1 & ,1 < t < 2 \\ 1 & , 2 < t \end{cases}$$

76)

Write the **Laplace transform** of the following:

a) $\frac{d^5y}{dt^5} =$ [1marks]

b) $(t+1)U(t-2)$

77)

Evaluate the following

$\mathcal{L}[e^{-4t}\delta(t-3)]$

$b)$ $\int_{-\infty}^{\infty} sin2t\delta(t-\frac{\pi}{4})dt$

78)

Solve the initial value problem $y'' - y = F(t)$ if $y(0) = 0$ and $y'(0) = 0$ where

$$F(t) = \begin{cases} 1, & 0 \le t < 1 \\ -1, & t \ge 1 \end{cases}$$ [10 marks]

79)

3. Solve using the method of separation of variables

$$\frac{\partial u}{\partial x} + \frac{\partial u}{\partial y} = 3u \quad \text{where } u(0,y) = e^{-5y}$$ [10 marks]

80)

Solve the initial value problem

$$y'' + 4y' + 13y = \delta(t-\pi), \quad \text{where } y(0) = 2 \text{ and } y'(0) = 1.$$ [10 marks]

81)

Find the Fourier series of the function

$$f(x) = \begin{cases} -1\,, & -\pi < x < -\frac{\pi}{2} \\ 0\,, & -\frac{\pi}{2} < x < \frac{\pi}{2} \\ 1, & \frac{\pi}{2} < x < \pi \end{cases}$$ [6 marks]

Solve using the method of separation of variables

$$\frac{\partial u}{\partial x} = 2\frac{\partial u}{\partial t} + u \quad \text{where } u(x,0) = 6e^{-3x}$$ [5 marks]

MATERIALS

MATH 204, TASK1, L200, REGULAR,202(

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3000 Solved Problems in Calculu

ES,S2,L200,MATH 204,2019.pdf

2018,Sem1,ES.pdf

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